Software Requirements Engineering

Lecture 2: Software Engineering and Software Requirement Engineering

Review

- Software requirement is a specification of what should be implemented.
- The importance of software requirements
- Software requirement engineering: requirement development and management

Outline

1. Software development models
2. Requirements engineering and software development
3. Software risk management

Software Development Model

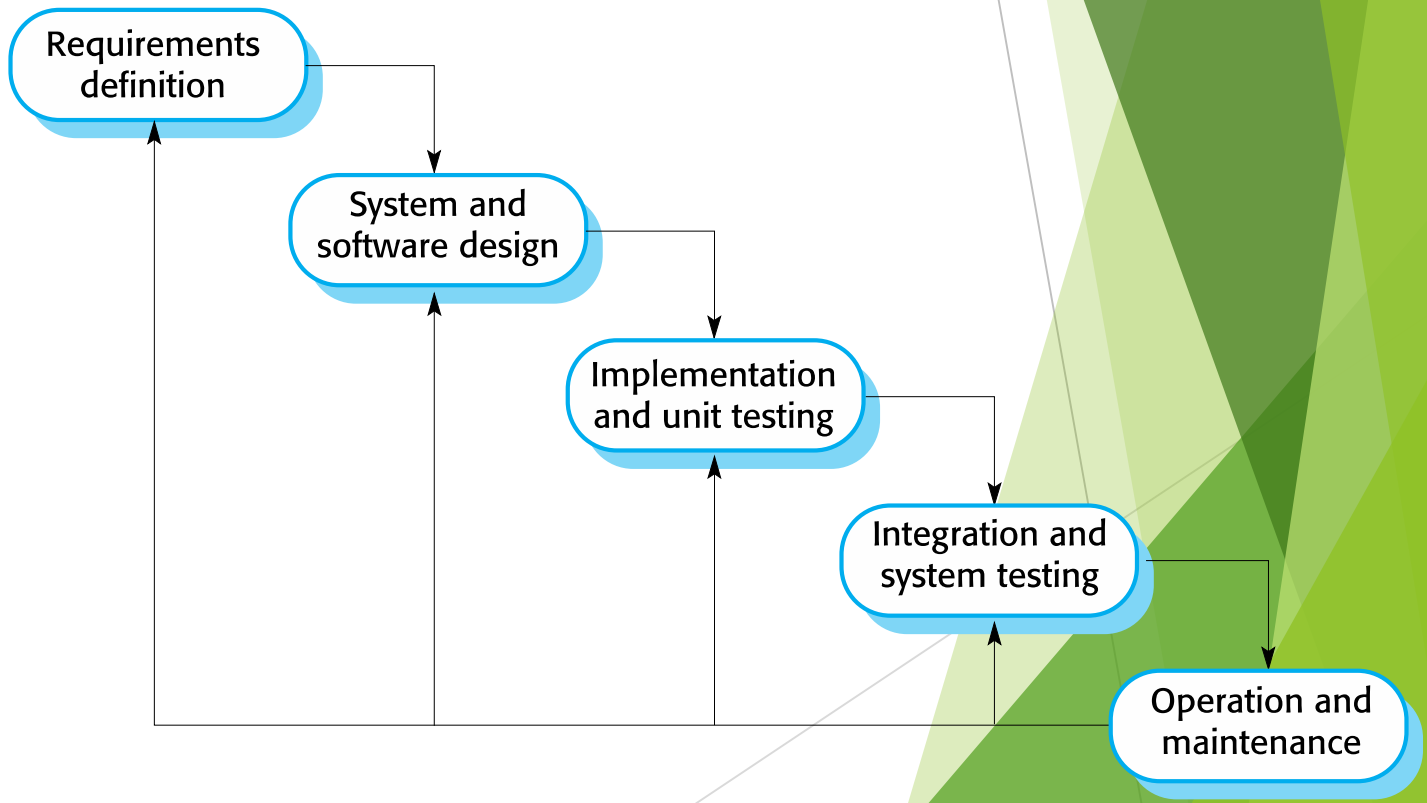
- The software process: A structured set of activities required to develop a software system.
- Many different software processes but all involve:
  - Specification – defining what the system should do;
  - Design and implementation – defining the organization of the system and implementing the system;
  - Validation – checking that it does what the customer wants;
  - Evolution – changing the system in response to changing customer needs.
- A software process model is an abstract representation of a process. It presents a description of a process from some particular perspective.

Software process models

- The waterfall model
  - Plan-driven model. Separate and distinct phases of specification and development.
- Incremental development
  - Specification, development and validation are interleaved. May be plan-driven or agile.
- Integration and configuration
  - The system is assembled from existing configurable components. May be plan-driven or agile.
- In practice, most large systems are developed using a process that incorporates elements from all of these models.

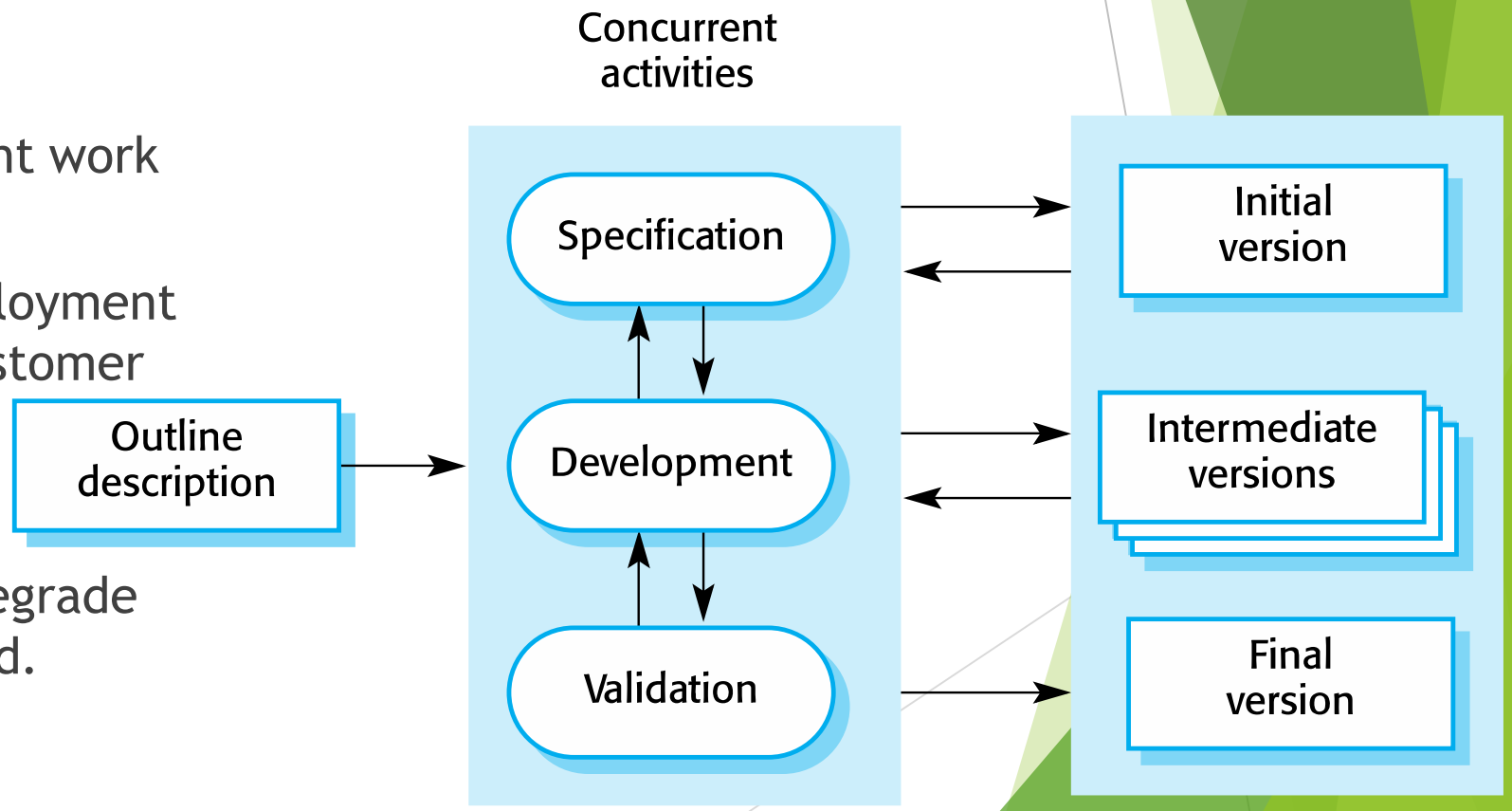
The Waterfall Model

- There are separate identified phases in the waterfall model:
  - Requirements analysis and definition
  - System and software design
  - Implementation and unit testing
  - Integration and system testing
  - Operation and maintenance
- The main drawback of the waterfall model is the difficulty of accommodating change after the process is underway. In principle, a phase has to be complete before moving onto the next phase.



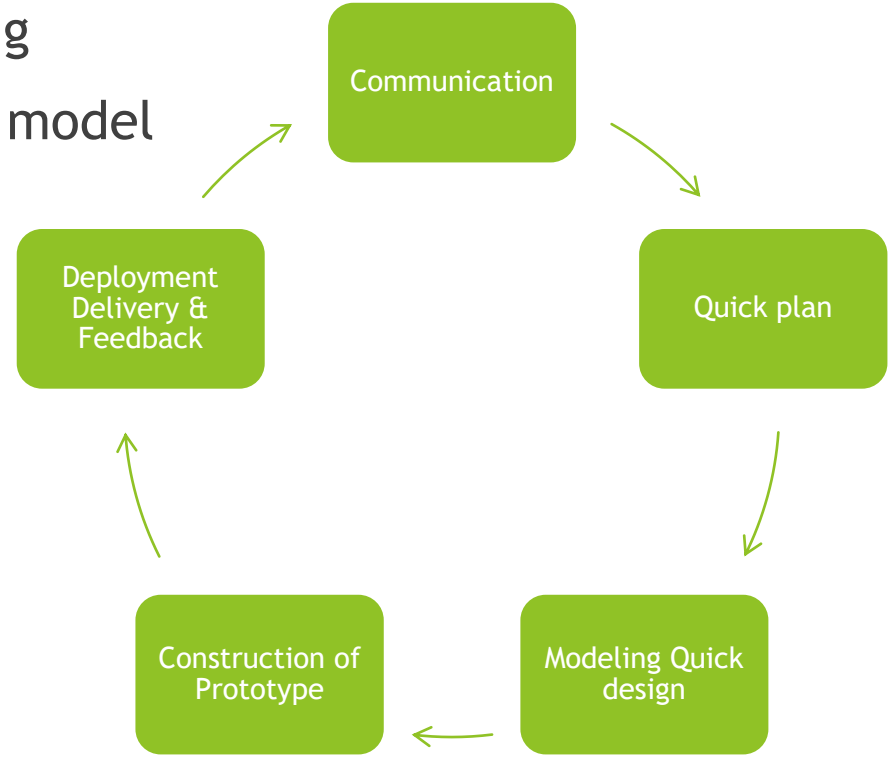
Incremental Process Model

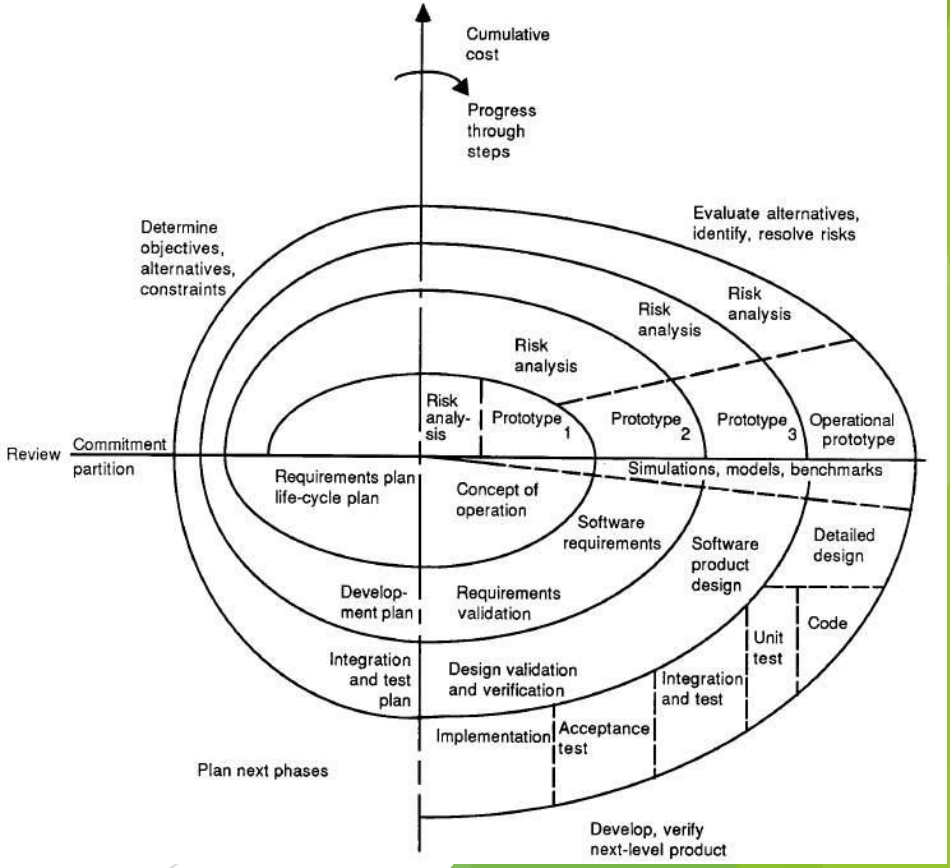
- The cost of accommodating changing customer requirements is reduced.
- It is easier to get customer feedback on the development work that has been done.
- More rapid delivery and deployment of useful software to the customer is possible.
- The process is not visible.
- System structure tends to degrade as new increments are added.



Evolutionary Process Model

- Evolutionary models are iterative. They are characterized in a manner that enables you to develop increasingly more complete versions of the software.
- Two common evolutionary process models:
  - Prototyping
  - The spiral model





Concurrent Models

- The concurrent development model, sometimes called concurrent engineering, allows a software team to represent iterative and concurrent elements of any of the other process models.
- Concurrent modeling defines a series of events that will trigger transitions from state to state for each of the software engineering activities, actions, or tasks.
- Concurrent modeling is applicable to all types of software development and provides an accurate picture of the current state of a project.

Software process models

- Change is inevitable in all large software projects. The system requirements change as the business procuring the system responds to external pressures and management priorities change. Therefore, whatever software process model is used, it is essential that it can accommodate changes to the software being developed.

2.3 Software Requirement Engineering and Software Development

- From the different software process models we can see that requirement engineering is an important stage in the software development process.
- Requirement engineering tasks are conducted to establish a solid foundation for design and construction.
- Requirements engineering occurs during the communication and modeling activities.

Software Requirement Engineering

- The influence of Requirement Engineering (RE) to software development:
  - Requirement is the basis of setting up plans for the projects.
  - The software requirement documentation is the fundamentals of software design and software implementation.
  - The software requirement document provides the basis of testing and validating.
  - The software maintenance should be done according to the software requirement document.

Requirements management

- The object of requirements management is to anticipate and accommodate the very real changes that you can always expect so as to minimize their disruptive impact on the project

2.3 Software Risk Management

- A software project group gathered for a discussion about their new project, a Chemical Tracking System. They all remember the problems they ran into on an earlier project called the Pharm Simulator.

Remember how we didn't find out that the users hated the Simulator's user interface until beta testing? It took us four weeks to rebuild it and retest it. I sure don't want to go through that death march again.

lead programmer

That was awful. It was also annoying that the users we talked to swore they needed a lot of features that no one has used so far. That drug interaction modeling feature took three times longer to code than we expected, and we wound up throwing it out. What a waste!

project manager

Maybe we should list these problems from the Simulator so we can try to avoid them on the Chemical Tracking System. I read an article on software risk management that said we should identify risks up front and figure out how to prevent them from hurting the project.

lead tester

I don't know about that. We probably won't have those same problems again. If we write down things that could go wrong on the Chemical Tracking System, it'll look like I don't know how to run a software project. I don't want any negative thinkers on this project. We have to plan for success!

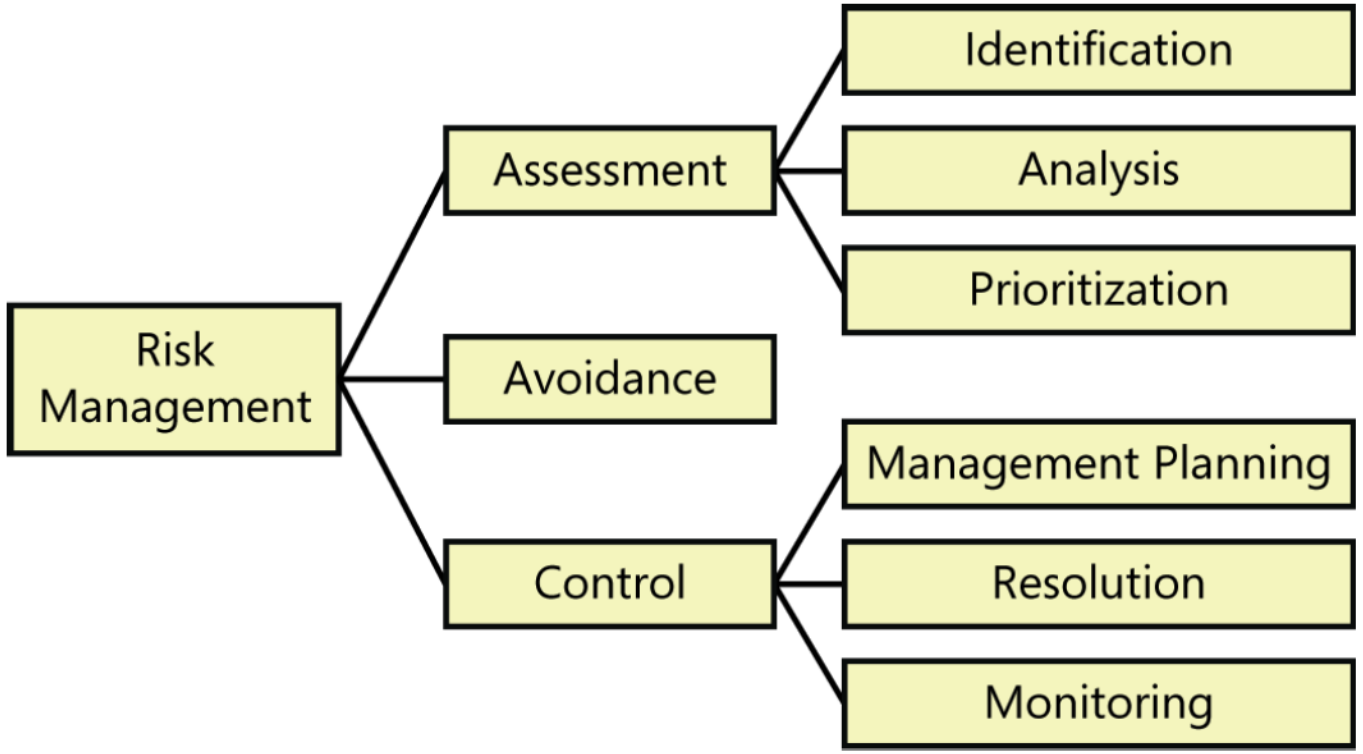
project manager

Software Risk

- A risk is a condition that could cause some loss or otherwise threaten the success of a project.
- Risk management is the process of identifying, evaluating, and controlling risks before they harm your project.
- Risk management involves the application of tools and procedures to contain project risk within acceptable limits.

Software Risk Management

Elements of risk management



Software Risk Management

- Risk assessment is the process of examining a project to identify potential threats.
- Risk avoidance is one way to deal with a risk: don't do the risky thing.
- Risk control activities are to manage the top-priority risks you identified.

Requirements-related Risks

- Requirements elicitation
  - Product vision and project scope
  - Time spent on requirements development
  - Customer engagement
  - Completeness and correctness of requirements specifications
  - Requirements for innovative products
  - Defining nonfunctional requirements
  - Customer agreement on requirements
  - Unstated requirements
  - Existing product used as the requirements reference
  - Solutions presented as needs
  - Distrust between the business and the development team

Requirements-related Risks

- Requirements analysis
  - Requirements prioritization
  - Technically difficult features
  - Unfamiliar technologies, methods, languages, tools, or hardware

Requirements-related Risks

- Requirements specification
  - Requirements understanding
  - Time pressure to proceed despite open issues
  - Ambiguous terminology
  - Design included in requirements

Requirements-related Risks

- Requirements validation
  - Unvalidated requirements
  - Inspection proficiency

Requirements-related Risks

- Requirements management
  - Changing requirements
  - Requirements change process
  - Unimplemented requirements
  - Expanding project scope

Software Risk Management

- Risk management helps you keep your eyes open and make informed decisions, even if you can’t control or avoid every adversity your project might encounter.

In the next lecture…

- We will talk about different types of requirements
- Use some examples to understand the concepts and difference between them

Question session